

AI Placement Prep Agent

Personalized Roadmap & Placement Preparation Powered by IBM watsonx Orchestrate

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1. Executive Summary & Problem Statement

The Challenge Students preparing for campus placements often rely on scattered resources, unverified advice, and generic study plans. They face significant ambiguity regarding: • Which core technical topics require maximum priority. • Which specific interview questions match their target job role. • How to convert current skill sets and available preparation timelines into an actionable, structured roadmap.	Project Objectives The primary goal is to provide an intelligent, personalized assistant that streamlines recruitment preparation: • Generate a week-by-week personalized preparation roadmap. • Provide role-specific technical and HR mock interview questions. • Conduct skill gap analysis against recruiter expectations. • Deliver an exportable preparation plan for offline study.
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2. Proposed Solution & Agentic Architecture

The solution features an interactive web interface powered by an embedded **IBM watsonx Orchestrate** AI agent. Rather than functioning as a static guidance page, the agent behaves as a goal-driven, context-aware preparation mentor.

Input Intake Captures student branch/major, current semester, target job role (e.g., SDE, Data Analyst), current comfortable skills, weak topics, and available preparation timeline.	Agentic Processing Processes profile information, maps individual competencies against industry expectations, and prioritizes learning topics based on impact.
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Action-Oriented Outputs

Generates concrete weekly schedules, prioritized topic matrices, curated interview question sets, and downloadable PDF preparation plans.

Interactive & Adaptive

Remains accessible via an embedded website widget, capable of refining recommendations dynamically through conversational follow-ups.

3. Key Technologies & Infrastructure

Component	Technology / Service	Description
AI Orchestration	IBM watsonx Orchestrate	Powers agent logic, conversational flow, profile analysis, and task orchestration.
Cloud Hosting	IBM Cloud	Provides deployment environment and infrastructure for the watsonx agent API integration.
Visual Workflow	IBM Bob Platform	Used for designing, orchestrating, and troubleshooting multi-agent workflows.
Frontend Interface	HTML5 / CSS3 / JavaScript (ES6+)	Responsive web application ('PlacementPrep AI') with embedded real-time chat interface.

4. Novelty, Key Differentiators & Future Scope

- **Hyper-Personalization:** Tailors prep plans directly to specific student profiles, time constraints, and targeted technical roles.
- **Skill-Gap Priority Focus:** Directly highlights and ranks top-priority topics (e.g., Advanced DSA, System Design, DBMS) required for interview clearance.
- **Multi-Channel Delivery:** Interactive web-agent experience backed by exportable offline PDF preparation roadmaps.

Future Development Roadmap:

1. *Resume Analysis:* Automated profile extraction via resume parsing.
2. *Adaptive Progress Tracking:* Dynamic schedule updating based on quiz performance.
3. *Voice Mock Interviews:* Real-time AI interview evaluations assessing speech, technical depth, and communication.

5. Completed IBM Certifications

- **Getting Started with Artificial Intelligence** (IBM SkillsBuild | Completion Date: 22 Aug 2026 | ID: PLAN-E624C2604060)
- **Lab: Troubleshoot Your Code Using IBM Bob** (IBM SkillsBuild | Completion Date: 22 Aug 2026 | ID: ALM-COURSE_4071307)